DATA SCIENCE (DSC) | INDIVIDUAL PROJECT

Retail Sales Forecasting

A Comparison of ARIMA, SARIMA and Holt-Winters Using an Interactive Power BI Dashboard

Mayur Chauhan | Enrollment No.: 230021713100 18
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Problem and data

- Problem: hard to see what sells, where, and what comes next.
- Data: public Sample Superstore orders, 9,994 lines (806 empty rows removed).
- Goal: clean, model, forecast with 3 methods, compare in Power BI.

9,994
order lines, 2015 to 2018

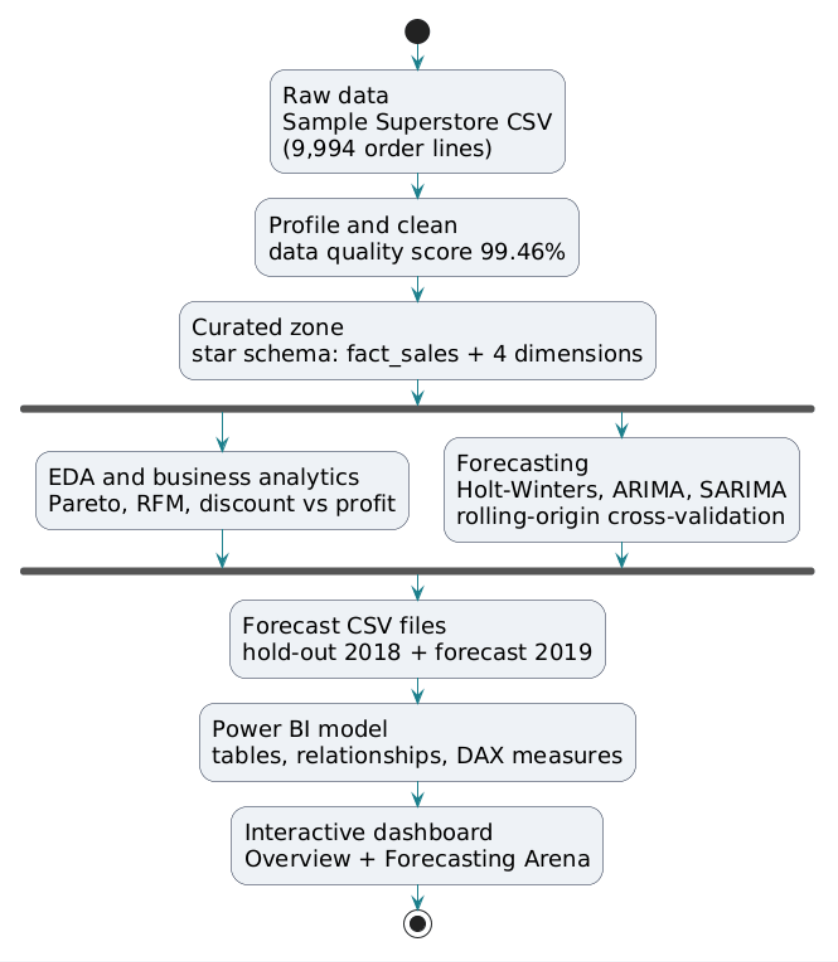
99.46%
data quality score

$2.30M
total sales

12.5%
overall profit margin

Real data, cleaned and checked before any model is fitted

Pipeline: from raw data to dashboard



1. Check quality: 99.46% score
2. Clean and fix types
3. Build the star schema
4. Explore: Pareto, RFM, discounts
5. Forecast with three methods and test them
6. Save CSV files
7. Power BI reads the CSV files

Every number in the dashboard comes from a script, so the whole pipeline can be re-run

Data model: star schema

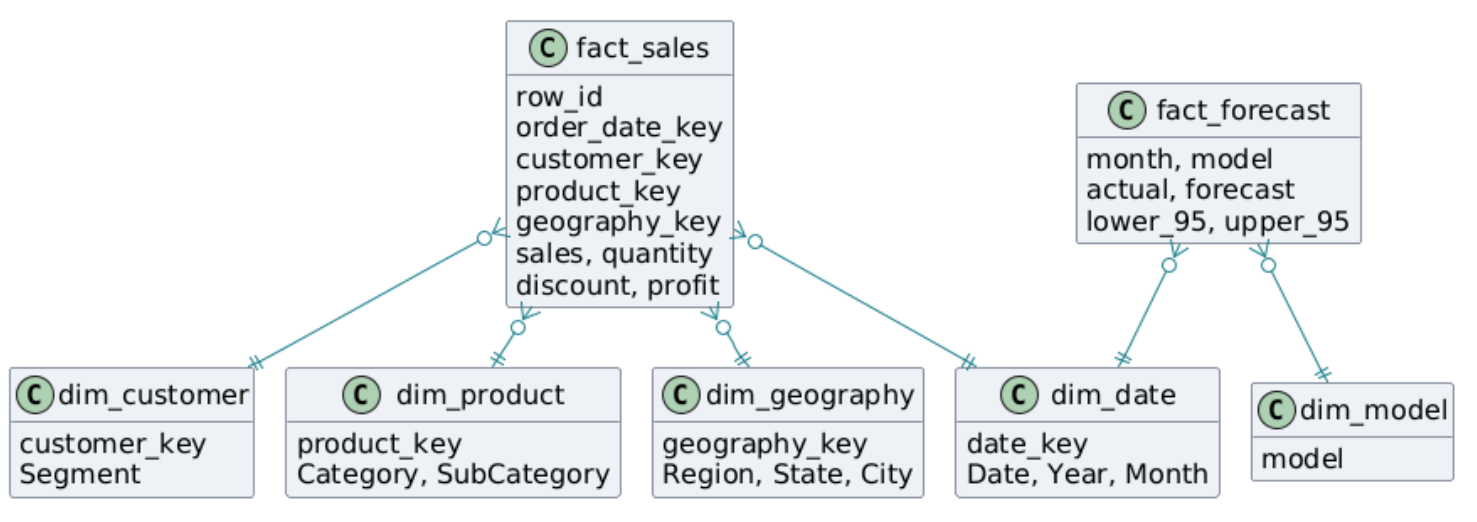
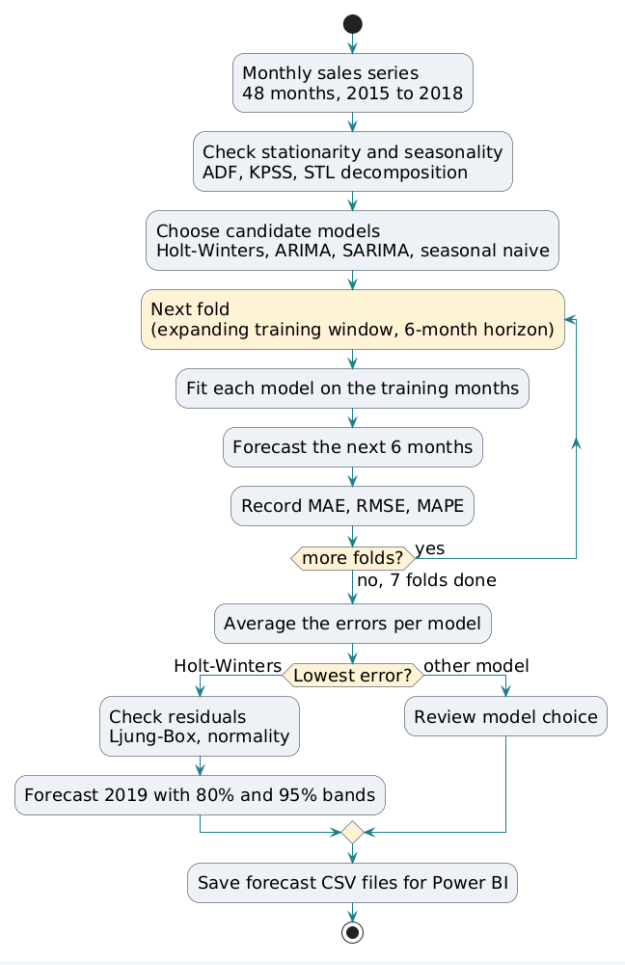


Table	Rows
fact_sales	9,994
dim_date	1,826
dim_product	1,862
dim_customer	793
dim_geography	632
fact_forecast	48
dim_model	3

- Fact table: one row per order line, with sales, quantity, discount and profit.
- Dimensions: date, product, customer and geography describe each sale.
- fact_forecast holds actual and forecast values by month and model; dim_model lets one slicer switch the model.

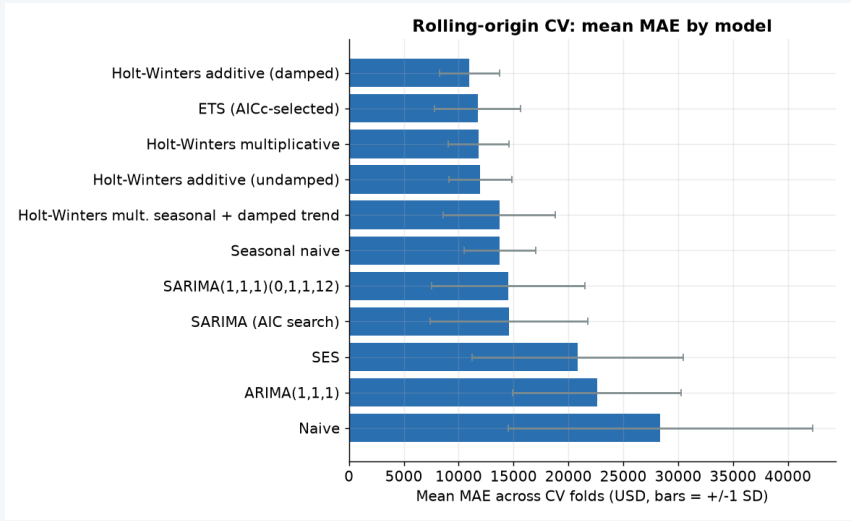
Standard star schema: fast to query and the shape Power BI expects

Forecasting: how the models were compared



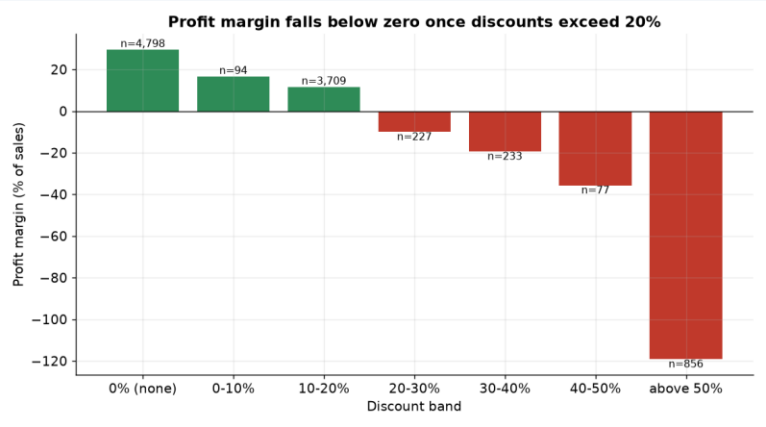
Model	CV MAE ($)	CV MAPE %	2018 MAPE %
Holt-Winters (damped)	**10,981**	**21.7**	**18.5**
Seasonal naive	13,739	25.0	24.5
SARIMA(1,1,1)(0,1,1,12)	14,513	36.6	34.0
ARIMA(1,1,1)	22,601	49.1	54.1

- CV = cross-validation: tested on 7 later windows of 6 months.
- MAE = average error in dollars.
- MAPE = average error in % of sales.
- 2018 MAPE = % error on the held-out 2018.
- Damped = the trend flattens out.
- ETS = automatic exponential smoothing.
- Lower is better.

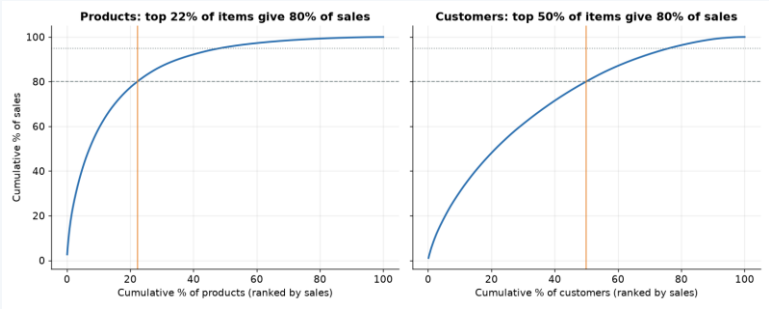


Holt-Winters has the lowest error, but its gap to SARIMA is not statistically significant (p = 0.375)

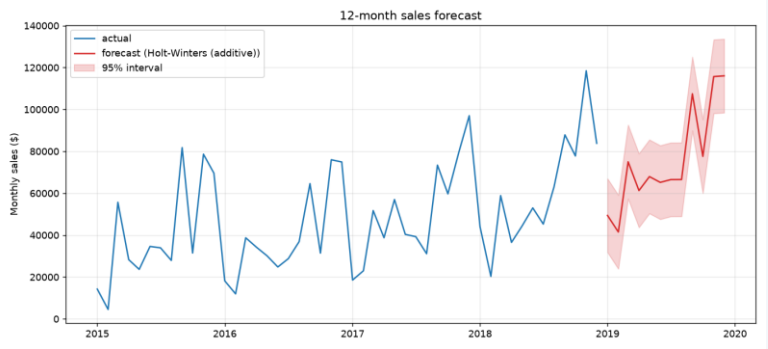
Findings



Discount vs profit



Pareto 80/20: few products, most sales

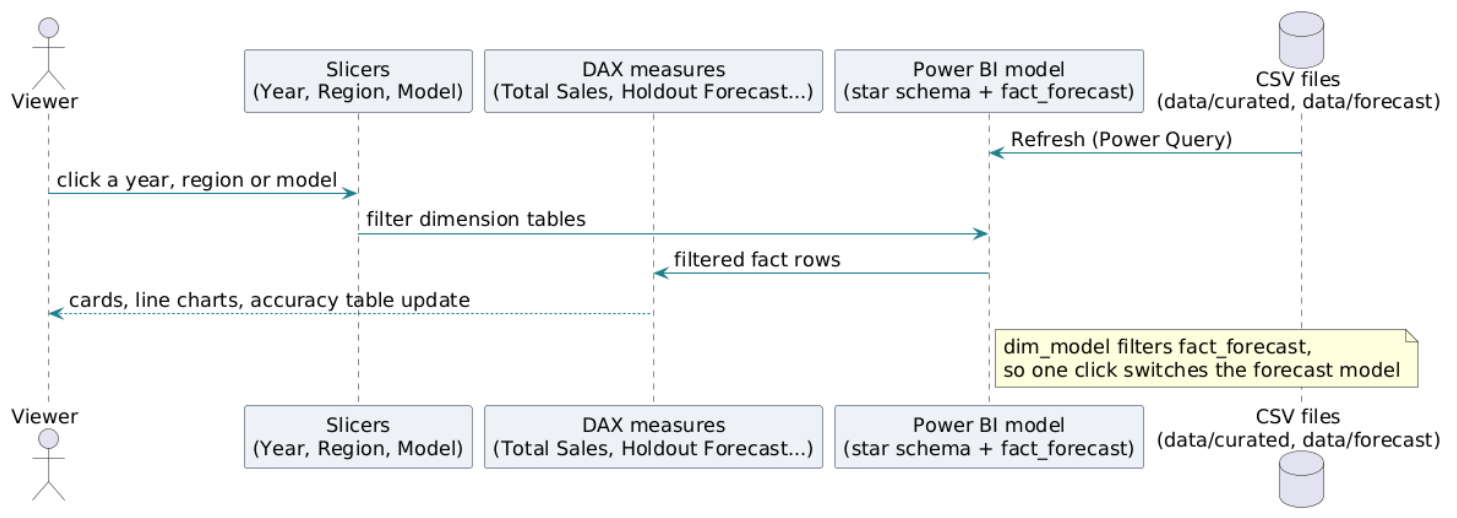


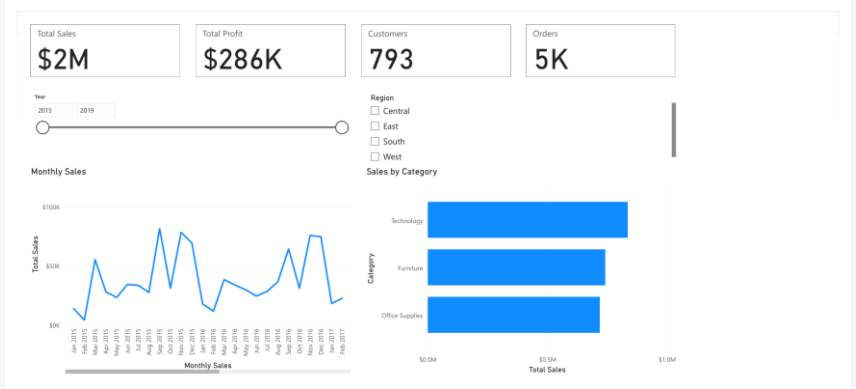
2019 sales forecast

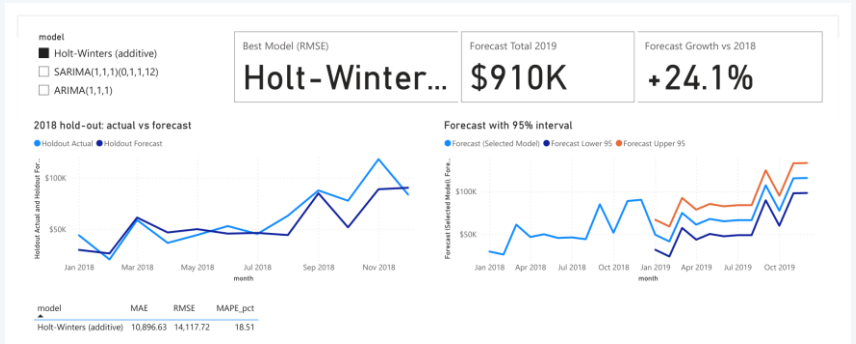
- Margin is 29.5% with no discount and 11.6% at 10 to 20% discount; lines above 20% discount lose 135,376 in total.
- Pareto 80/20 rule: 22.2% of products (414 of 1,862) give 80% of sales.
- Holt-Winters forecasts about $910K for 2019.

Discounting is the main profit risk; Holt-Winters is the best forecaster here

Power BI dashboard and conclusion







- Built: Overview and Forecasting Arena pages, with Year, Region and Model slicers.
- Limits: 48 months of data, overlapping folds, two pages, the 2019 forecast covers the best model only.
- Conclusion: Holt-Winters forecasts retail sales best here, and the dashboard lets users compare the three methods.

Answer to the title: Holt-Winters is the most accurate of the three methods on this data